Lecture 1: Vector Spaces and Subspaces

A vector space is a set V, together with two operations (vector addition and scalar multiplication), satisfying eight axioms: closure under addition, associativity of addition, commutativity of addition, existence of a zero vector, existence of additive inverses, closure under scalar multiplication, distributivity of scalar multiplication over vector addition, and distributivity of scalar multiplication over field addition.

Common examples include R^n (n-tuples of real numbers), the space of polynomials of degree at most n, and spaces of continuous functions on an interval. A subspace is a subset of a vector space that is itself a vector space under the inherited operations; to check whether a subset W of V is a subspace, it suffices to verify that W contains the zero vector and is closed under both addition and scalar multiplication.

Lecture 2: Linear Independence, Basis, and Dimension

A set of vectors {v1, ..., vk} is linearly independent if the only solution to c1*v1 + c2*v2 + ... + ck*vk = 0 is c1 = c2 = ... = ck = 0. If a nontrivial combination equals zero, the set is linearly dependent, meaning at least one vector can be written as a combination of the others.

A basis for a vector space V is a linearly independent set that spans V   every vector in V can be written uniquely as a linear combination of basis vectors. All bases of a given finite-dimensional vector space have the same number of elements, called the dimension of V, written dim(V).

Lecture 3: Matrices and Linear Transformations

A linear transformation T: V -> W between vector spaces preserves vector addition and scalar multiplication: T(u + v) = T(u) + T(v) and T(c*v) = c*T(v). Every linear transformation between finite-dimensional vector spaces can be represented by a matrix once bases for V and W are fixed.

Composing two linear transformations corresponds exactly to multiplying their matrix representations. The rank of a matrix is the dimension of its column space (equivalently, the dimension of the image of the transformation), and the rank-nullity theorem relates rank, nullity, and the dimension of the domain: dim(domain) = rank(T) + nullity(T).

Lecture 4: Eigenvalues, Eigenvectors, and Diagonalization

For a square matrix A, a nonzero vector v is an eigenvector of A if Av = lambda*v for some scalar lambda, called the corresponding eigenvalue. Eigenvalues are found as the roots of the characteristic polynomial, det(A - lambda*I) = 0, and the eigenvectors for a given eigenvalue span its eigenspace.

A matrix A is diagonalizable if it can be written as A = P*D*P^-1, where D is diagonal and P’s columns are linearly independent eigenvectors of A. Diagonalization is possible exactly when the algebraic and geometric multiplicities of every eigenvalue match, and it dramatically simplifies computing powers of A (A^k = P*D^k*P^-1), which underlies methods like Principal Component Analysis and the power iteration method.